

House Price Prediction

Internship Project Report

Mohammad Zuheer

Xylofy AI Internship

June 2026

Objective

- Build a machine learning model to predict house prices.
- Identify the most influential features affecting house prices.
- Compare the performance of Linear Regression and Random Forest Regressor.

Dataset Information

Attribute	Details
Dataset	Housing Prices Dataset (Kaggle)
Total Records	545
Target Variable	Price
Features	Area, Bedrooms, Bathrooms, Stories, Parking, Furnishing Status, Air Conditioning, Basement, Preferred Area, etc.

Data Preprocessing

The dataset was loaded using Pandas and inspected for missing values and duplicate records. No missing values were found. Categorical variables were converted into numerical representations using one-hot encoding. Finally, the dataset was divided into training and testing sets using an 80:20 split.

Exploratory Data Analysis

Initial analysis showed that house prices follow a slightly right-skewed distribution. Property area exhibited the strongest positive relationship with price, while bathrooms, stories, parking spaces, and furnishing status also demonstrated noticeable influence. Furnished houses generally had higher prices than semi-furnished and unfurnished properties.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.6529
Random Forest	1,013,969	1,398,116	0.6132

Table 1: Comparison of Machine Learning Models

Linear Regression achieved better predictive performance than Random Forest across all evaluation metrics. The lower MAE and RMSE values indicate more accurate predictions, while the higher R² score suggests better overall explanatory power.

Feature Importance

Feature Importance Plot Generated Using Random Forest Regressor

The feature importance analysis revealed that **Area** is the most influential factor affecting house prices. Other important contributors include the number of bathrooms, parking availability, air conditioning, and the number of stories.

Key Findings

- Area was the strongest predictor of house price.
- Bathrooms, parking, and air conditioning significantly increased property value.
- Furnishing status had a noticeable effect on market price.
- Linear Regression outperformed Random Forest because the dataset exhibited a predominantly linear relationship between features and house prices.

Conclusion

This project successfully developed machine learning models for house price prediction. Linear Regression produced the best results with an R² score of 0.6529 and an average prediction error of approximately Rs. 9.7 lakh. The study highlights that property area and essential amenities are the primary drivers of housing prices, making them critical factors for valuation and decision-making.

End of Report